

Krish Naik Course Roadmap

Module / Topic Breakdown per Course (CampusX DSMP excluded)

1. Complete Python With DSA Bootcamp + LEETCODE Exercises

Focus: Python fundamentals + Data Structures & Algorithms + interview coding practice

- Python syntax, data types, control structures, functions
- Core data structures: arrays, strings, linked lists, stacks, queues
- Trees, binary search trees, heaps
- Sorting algorithms: selection, bubble, insertion, merge, quick
- Searching algorithms: linear, binary, ternary search
- Recursion and backtracking basics
- LeetCode problem-solving practice (easy to interview-level)
- Time and space complexity analysis
- Practical app-building project applying Python + data structures
- Interview preparation cheat sheets and technical interview readiness

2. Mathematics - Basics to Advanced for Data Science and GenAI

Focus: Math foundations underlying ML/DL algorithms

- Linear Algebra: vectors, matrices, linear transformations, eigenvalues/eigenvectors
- Linear Algebra applications: PCA, dimensionality reduction, neural network foundations
- Descriptive statistics: mean, median, variance, distributions
- Inferential statistics: hypothesis testing, confidence intervals
- Probability theory: fundamentals and applications in ML
- Differential calculus: derivatives, gradients
- Integral calculus and optimization (gradient descent foundations)
- Application of math concepts to regression, neural networks, and deep learning

3. Complete Generative AI Course With Langchain and Huggingface

56 sections | 258 lectures | ~70 hours

- Python refresher: syntax, OOP, file handling, exception handling
- Streamlit basics for building ML/AI web apps
- NLP prerequisites: tokenization, stemming, lemmatization, stopwords, POS tagging, NER
- Text vectorization: One-Hot Encoding, Bag of Words, TF-IDF, N-grams
- Word embeddings: Word2Vec (CBOW, Skip-gram), average word2vec
- Introduction to LangChain framework and its role in GenAI development
- Huggingface model integration into LangChain projects
- Fine-tuning and customizing Huggingface models
- Building Retrieval-Augmented Generation (RAG) pipelines
- Deployment strategies: cloud platforms and on-premise servers
- Real-world projects: chatbots, content generators, data augmentation tools
- Monitoring and optimizing deployed GenAI applications

4. Complete MLOps Bootcamp With 10+ End-to-End ML Projects

24 sections | 180 lectures | ~51 hours

- Python prerequisites: syntax, OOP, logging, exception handling
- IDE setup: Google Colab, GitHub Codespace, Anaconda + VS Code
- Flask framework: routing, templates, REST APIs (GET/POST/PUT/DELETE)
- Git and GitHub: version control, branching, merge conflict resolution
- MLflow: experiment tracking, model registry, tracking server
- ML project integration with MLflow (e.g. house price prediction)
- Deep Learning (ANN) integration with MLflow tracking
- DVC (Data Version Control) for dataset and model versioning
- DagsHub: remote MLflow-backed repository, team collaboration
- Docker containerization for ML models
- Apache Airflow with Astronomer: ETL pipeline automation
- CI/CD pipelines using GitHub Actions
- End-to-end ML project: data collection to deployment
- End-to-end NLP project deployment and monitoring with Huggingface
- AWS SageMaker: model deployment, scaling, monitoring
- Generative AI deployment using AWS cloud infrastructure
- Monitoring with Grafana dashboards connected to PostgreSQL

5. AI Security Bootcamp - Guardrails, LLM Gateways, Observability

31 sections | 261 lectures | ~75 hours

- LangChain fundamentals: agents, tools, structured output, middleware
- Human-in-the-loop middleware for high-stakes agent actions
- Building AI agents and chatbots with LangGraph
- Multi-agent systems: simple, supervised, hierarchical architectures
- MCP (Model Context Protocol) server/client implementation
- AI observability fundamentals: logging, tracing, metrics
- Observability with LangSmith: tracing, RAG tracing, agentic workflows
- Observability with Pydantic Logfire: structured logging, RAG/agent tracing
- NVIDIA NeMo Guardrails: Colang rails, input/output rails, PII detection
- Guardrails AI: validators, on-fail actions, structured output validation
- AWS Bedrock Guardrails: content filters, denied topics, PII redaction, grounding checks
- LLM Gateways: Portkey AI Gateway and TensorZero (routing, fallback, caching, rate limiting)
- LLM evaluation using DeepEval (hallucination, faithfulness, custom metrics)
- AI Red Teaming with PyRIT (prompt injection, jailbreak defense)
- Redis for semantic caching and performance optimization
- Building Deep Agents and AI coding assistants with Claude Code

6. Big Data Engineering Bootcamp with GCP and Azure Cloud

Zero-to-expert big data engineering track

- Big Data fundamentals: the 5 V's (Volume, Velocity, Variety, Veracity, Value)
- SQL/MySQL: database and table creation, queries, joins, data manipulation
- Python scripting fundamentals for data engineering
- Hadoop ecosystem for distributed storage and processing
- Apache Spark: batch processing, distributed computing
- Apache Kafka: real-time and streaming data pipelines
- Apache Flink: stream processing
- Batch vs near-real-time vs real-time processing architectures
- Cloud platforms: AWS, Azure, and GCP for data engineering
- Building scalable, end-to-end data pipelines
- Real-world end-to-end big data projects